

SUVAB BARAL

SOFTWARE ENGINEER

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WORK EXPERIENCE:

Software Engineer II Trimble Inc, USA 10/2024 – Present

- Design and develop end-to-end features for Trimble Fleet Maintenance Platform, leading development across a complex, **multi-layered codebase starting from the solution architecture, UI/UX, REST APIs to database design**.
- Collaborated in an Agile environment to understand customer requirements and built quality features including Single Sign-On across Trimble products, interactive dashboards, and order/inventory/technician management modules.
- Strong focus on integrating AI tools and features, enhancing both product capability and engineering efficiency.

Software Engineer and Research Assistant Research Foundation for SUNY, USA 08/2023 – 12/2024

- Developed a scalable and secure **infrastructure for continuous health monitoring** system using Raspberry Pi, Node.js, Python, MQTT, MySQL, InfluxDB, and AWS IoT for storing and analyzing continuous time series data from Ultra wide-band (UWB) sensors aiding in vital signs monitoring and early detection of symptoms.
- Analyzed various machine learning models, including **LLMs and transformers, to optimize activity recognition**, location trajectory, and vital signs monitoring to develop **a mobile app for home self-assessment** of vital signs.

Associate Technical Lead Rently, India 03/2023 – 08/2023

- Led the architecture **design and engineering of Rently's Monthly Recurring Revenue Billing System** by designing the overall billing experience and integrating Netsuite 360SB.
- Achieved a **35% reduction in manual billing workload**, alongside a **30% decrease in billing and invoicing issues**.

Senior Software Engineer Rently, India 03/2022 – 03/2023

- Designed a **IoT device monitoring system** for Rently's smart locks and thermostats using Node.js, Java, AWS EC2, DynamoDB, PostgreSQL, MQTT, and Apache Kafka.
- Developed a data analysis algorithm to enhance security, monitoring, and energy conservation from IoT device data.
- Improved the algorithm performance** in incremental releases **by 50%** through multithreading and parallel processing which helped achieved **over 70% adoption rate** for the monitoring system.
- Increased team velocity by 2-4 story points per sprint** as a Scrum Master, and helped in talent acquisition as technical reviewer, recruiter, and mentor.

Software Engineer Rently, India 09/2019 – 03/2022

- Mastered full-stack agile development, creating Rently's cross-platform smart home app in React Native and server-side solutions with Node.js, Java, Python, PostgreSQL, and AWS Lambda/EC2/Cognito/DynamoDB.
- Developed the **Bluetooth library and Rently Smart Home Alexa skill** for seamless communication with smart devices.
- Maintained over **95% test case coverage** for all projects following Test Driven Development (TDD).

EDUCATION:

Stony Brook University, Master of Science (M.S) – Computer Engineering 2024
CGPA: 3.83

Sri Ramakrishna Engineering College, Bachelor of Engineering (B.E) – Computer Science & Engineering 2020
CGPA: 3.8, Gold medalist for Computer Science and Engineering department.

SKILLS:

- Programming Languages:** Node.js, Java, Python, C, C++, C# (.NET framework), React.js, React Native, Angular.
- Tools:** Linux, Git, Jira, AWS, Azure, Android Studio, Serverless, MQTT, Docker, Kubernetes, Raspberry Pi, ApacheKafka, ORMs, Tensorflow, Keras, CUDA, Bluetooth technology, Postman, Jenkins.
- Databases:** PostgreSQL, MSSQL, AWS Dynamo DB, MongoDB, Redis, InfluxDB.
- Languages:** English, Nepali, Hindi.
- Skilled in designing robust backend servers, REST APIs and cross-platform mobile app with automated unit testing.
- Proficient in machine learning models and Large Language Model (LLM) integration.
- Proficient with CI/CD tools, including Convex, Jenkins and CircleCI.
- Experienced in time series analysis and data ETL pipelines with Apache Kafka.

PUBLICATION:

Proteus: An Easily Managed Home-Based Health Monitoring Infrastructure. <https://doi.org/10.1109/JIOT.2024.3487875>